

Transformation of Random Variables

Definition

The objective is to find the Probability Density Function (PDF) or Cumulative Distribution Function (CDF) of a new random variable (RV), denoted as Y , which is a function of an existing random variable X (i.e., $y = g(x)$). The PDF $f_X(x)$ and CDF $F_X(x)$ of X are known a priori.

1 Transformation of Random Variables ($y = g(x)$)

Assumptions

- $y = g(x)$ is a transformation function.
- The inverse function $x = g^{-1}(y)$ exists.

Key Results

- **Step 1 (CDF Method):** Find the CDF of Y : $F_Y(y) = P(Y \leq y)$.
- **Step 2 (PDF Derivation):** Differentiate the CDF with respect to y to find $f_Y(y)$.

Derivation

Step 1:

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) \\ &= \Pr(g(X) \leq y) \\ &= \Pr(X \leq g^{-1}(y)) \\ &= F_X(g^{-1}(y)). \end{aligned} \tag{1}$$

Step 2:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))] = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y). \tag{2}$$

General Formula:

$$\boxed{f_Y(y) = \left. \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \right|_{x=g^{-1}(y)}} \tag{3}$$

Example

Problem: X is a uniformly distributed RV on $(-1, 1)$. Find the PDF of $Y = \sin\left(\frac{\pi X}{2}\right)$.

Solution:

1. **Given PDF of X :**

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

2. **Transformation:**

$$y = \sin\left(\frac{\pi x}{2}\right) \implies x = \frac{2}{\pi} \sin^{-1}(y).$$

3. **Derivative:**

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}.$$

4. **Final PDF:**

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right| = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}} = \frac{1}{\pi \sqrt{1-y^2}}, \quad -1 < y < 1.$$

2 Function of Two Random Variables

Topic: Sum of Two Random Variables ($Z = X + Y$)

Assumptions

- X and Y are independent random variables.

Key Result: Convolution Integral

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x) dx. \quad (4)$$

Derivation

1. **CDF Setup:**

$$F_Z(z) = \Pr(Z \leq z) = \Pr(X + Y \leq z).$$

2. **Double Integral:**

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy.$$

3. **Leibniz Rule Application:**

To find $f_Z(z) = \frac{d}{dz} F_Z(z)$, apply the Leibniz rule:

$$\frac{d}{dz} \int_{a(x)}^{b(x)} h(x, y) dy = h(x, b(x)) \frac{db}{dz} - h(x, a(x)) \frac{da}{dz} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y) dy.$$

4. **Result:**

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z-y, y) dy.$$

For independent X and Y , this simplifies to the convolution of f_X and f_Y .

Illustrative Examples

Example 1: Normal Distribution

- **Given:** $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$, independent.
- **Goal:** Prove $Z = X + Y \sim \mathcal{N}(0, 2)$.
- **Setup:**

$$f_Z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-x^2/2} \cdot e^{-(z-x)^2/2} dx.$$

- **Conclusion:** After integration, $Z \sim \mathcal{N}(0, \sigma^2)$ where $\sigma^2 = 2$.

Example 2: Exponential Distribution

- **Given:** X and Y are exponential RVs with parameter λ .
- **Setup:**

$$f_Z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} u(x) \cdot \lambda e^{-\lambda(z-x)} u(z-x) dx.$$

3 Other Transformations

- **Difference:** $Z = X - Y$.

$$F_Z(z) = \Pr(X - Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y) dx dy.$$

- **Ratio:** $Z = \frac{X}{Y}$.

$$F_Z(z) = \Pr\left(\frac{X}{Y} \leq z\right).$$

- **Magnitude:** $R = \sqrt{X^2 + Y^2}$.